

LECTURER INFORMATION

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KẾ HOẠCH GIẢNG DẠY

01



Kế hoạch học tập:

Phân bổ từng tuần theo đề cương môn học

02



Kiểm tra:

- Bài tập hàng tuần
- Bài thi thực hành
- Bài tập nhóm

03



Thi kết thúc học phần:

Đồ án môn học, làm theo nhóm, full stack web

PHƯƠNG PHÁP ĐÁNH GIÁ

01



Cách thức đánh giá:

- | | |
|------------------------------|-----|
| - Chuyên cần + Bài tập | 20% |
| - Giữa kỳ (Đồ án môn học L1) | 20% |
| - Cuối kỳ (Đồ án môn học L2) | 60% |

02



Điều kiện dự thi:

- Điểm giữa kỳ > 0 mới được dự thi cuối kỳ

03



Yêu cầu khác:

Áp dụng các tiêu chuẩn lập trình khi làm đồ án



PROJECT PRESENTATION

ShopHub: E-Commerce System

Full-Stack Development with ReactJS and Python FastAPI

Lecturer: Msc. Luong Tran Hy Hien

Executive Summary



Integrated Solution

ShopHub is a comprehensive Full-Stack E-Commerce application designed to bridge the gap between seamless consumer experiences and powerful administrative management. It leverages modern asynchronous processing and a decoupled architecture.

Key Deliverables

The system encompasses a high-performance product catalog, secure identity management, stateful shopping cart dynamics, and a robust order fulfillment lifecycle—all deployed via cloud-ready containerized workflows.

Seamless Customer Journey



Selection

Dynamic carting

Checkout

Order processing

Discovery

Fast search & filter

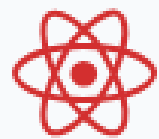
Identity

Secure JWT Auth

Fulfillment

History tracking

Frontend Tech Stack



ReactJS + Vite

Optimized build tooling for lightning-fast HMR and reactive component-based development.



React Router

Declarative navigation for a seamless single-page application (SPA) experience.



Material UI

Enterprise-grade design system ensuring a professional and responsive user interface.

Backend Tech Stack



FastAPI

High-performance Python framework utilizing modern asynchronous programming models.



PostgreSQL

ACID-compliant relational database for robust data integrity and complex relationship mapping.



SQLAlchemy

Powerful Object-Relational Mapping (ORM) to unify Python logic with database persistence.

Role-Based Access Control

ShopHub implements a stateless authentication model using JSON Web Tokens. Combined with **RBAC**, the system strictly separates customer privileges from administrative controls.

-  Secure password hashing
-  Admin vs Customer roles
-  Protected API endpoints

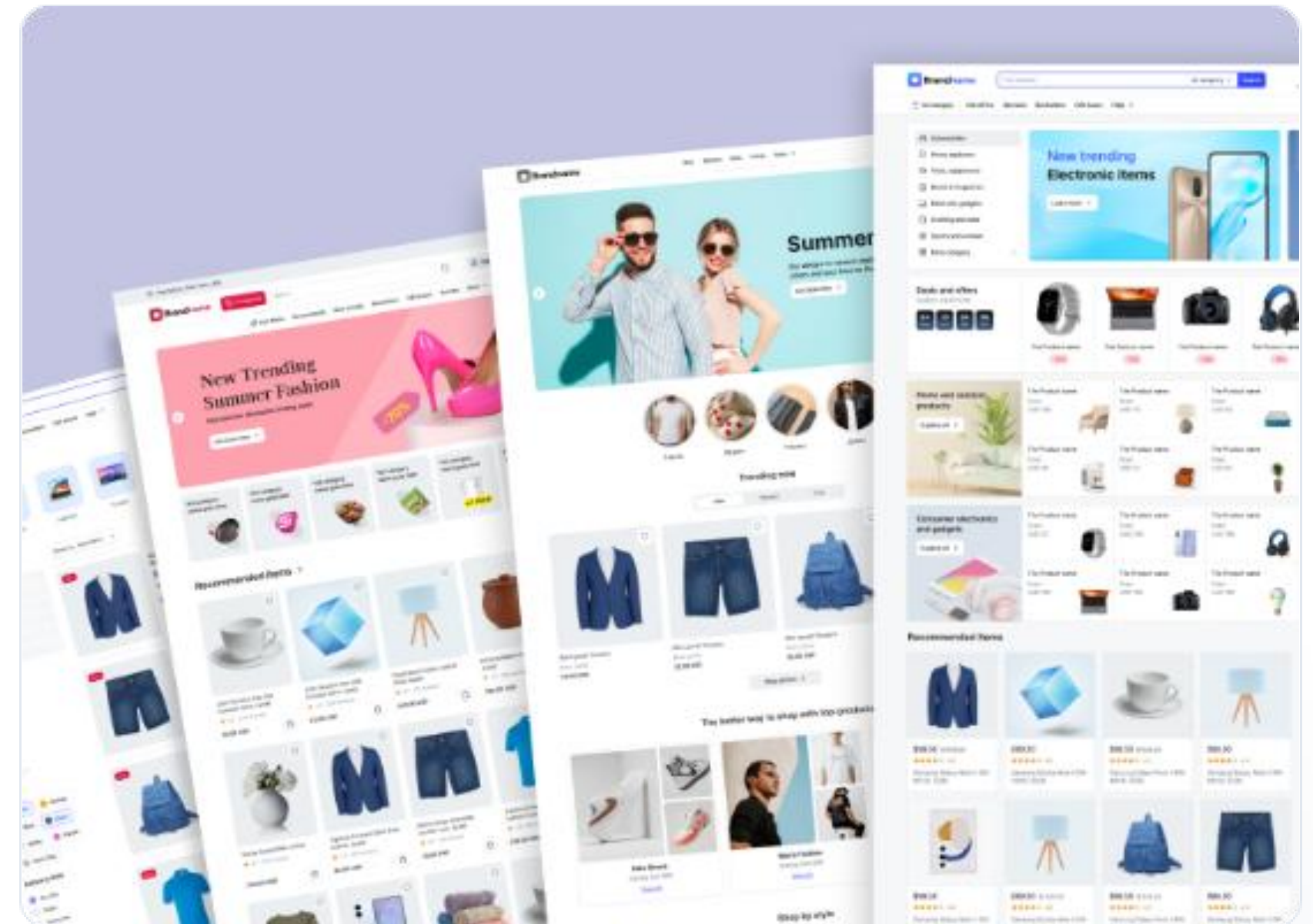
JWT

Product Discovery Experience

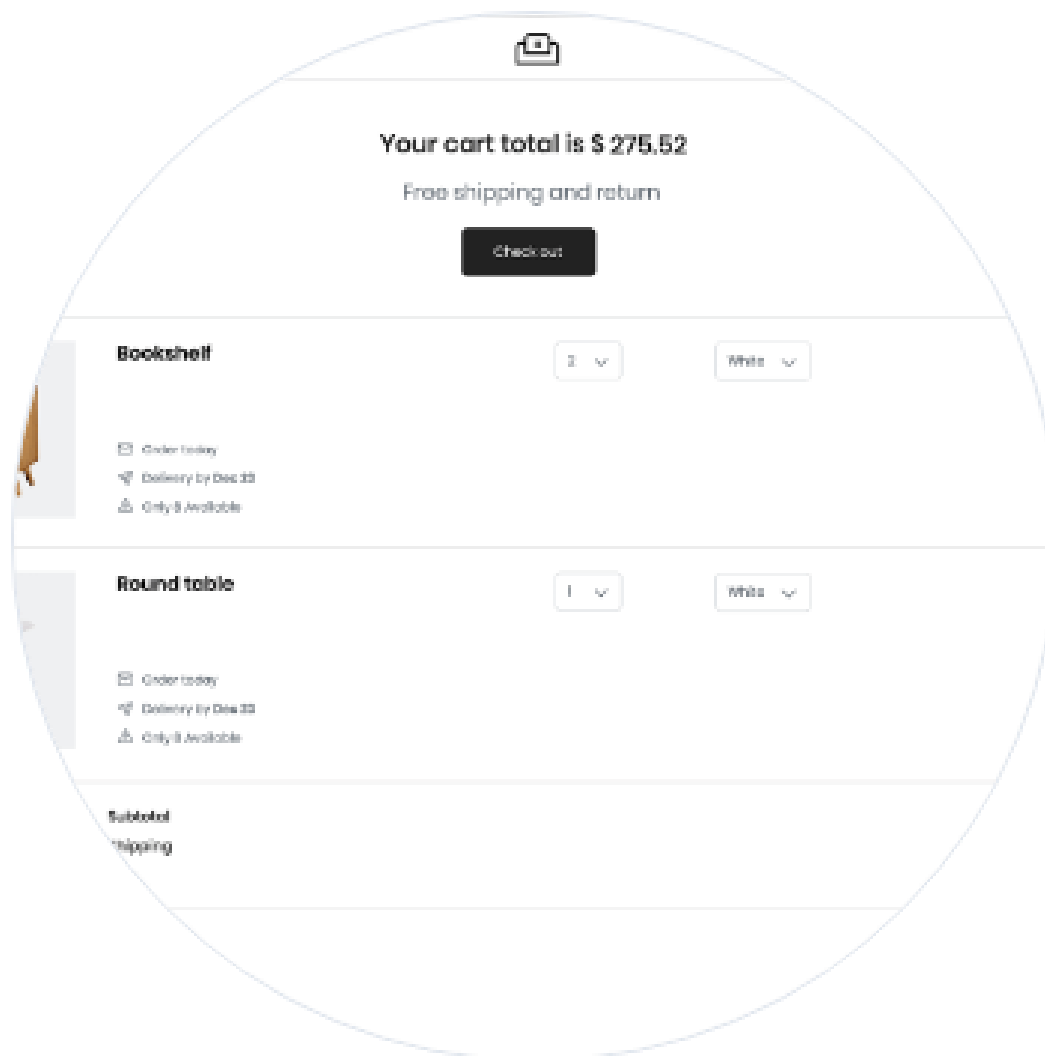
Interactive Catalog

The discovery layer is built for speed and clarity, featuring:

-  Real-time keyword filtering
-  Category-based navigation
-  Detailed specification views
-  Price and rating indicators



| Stateful Shopping Cart



Context-Driven Persistence

The shopping cart utilizes React Context API for global state management, ensuring that user selections remain consistent across different views. Integrated local persistence allows users to return to their pending selections even after page refreshes.

Seamless transition from browsing to commitment.

Order Lifecycle Management



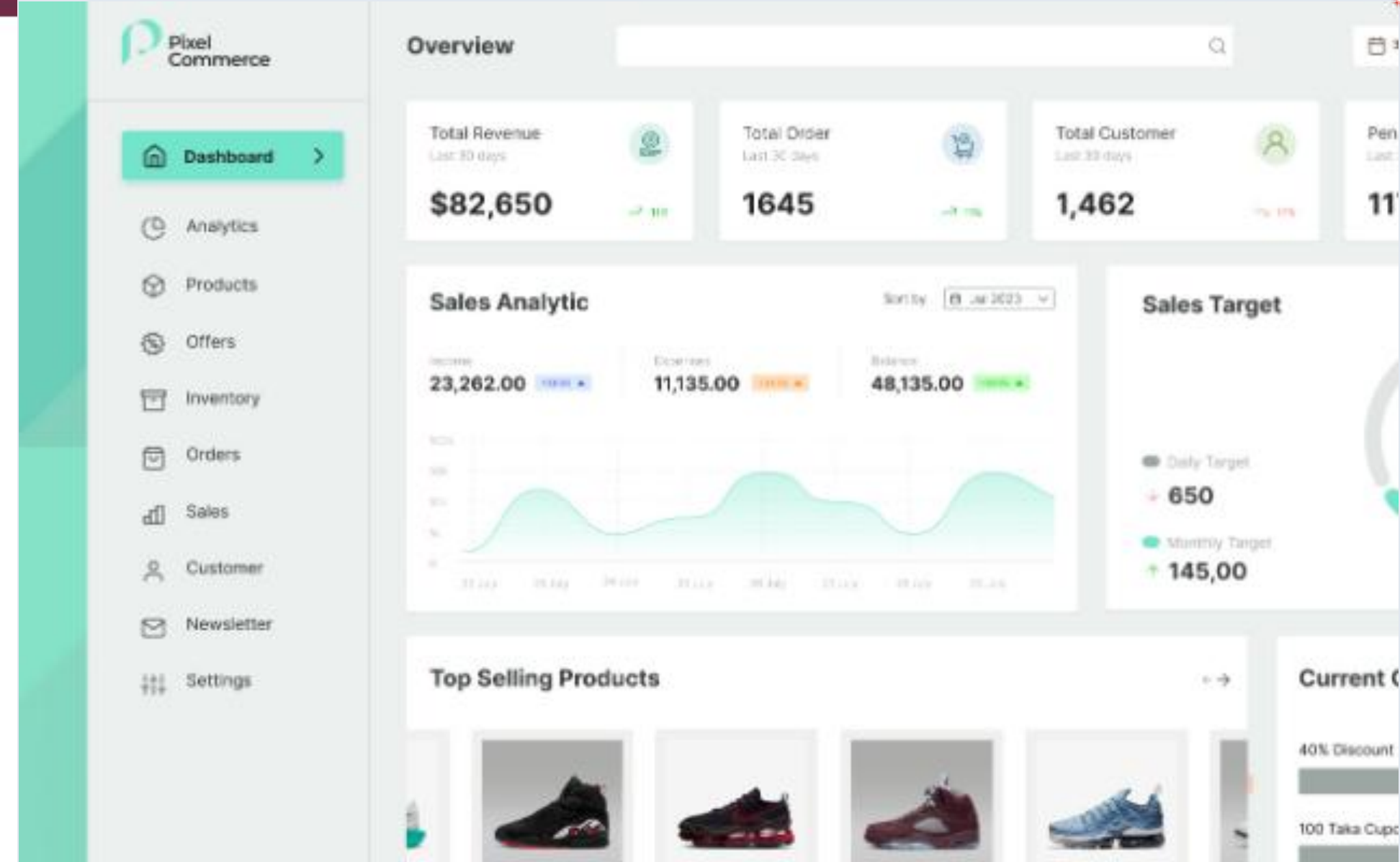
Workflow Phase	Backend Action	User Status
Checkout	Transaction validation	Order Placed
Processing	Inventory deduction	Payment Confirmed
Shipment	Carrier ID assignment	Dispatched
Completion	Final history archival	Delivered

Admin Command Center

Operational Control

Admins have access to a dedicated dashboard for total platform oversight. Manage your inventory, monitor sales trends, and update order statuses in a centralized interface designed for efficiency.

The dashboard provides real-time insights into system health and commercial performance.



Project timeline



- Sessions 1–5: Frontend basics – React, JSX, components, routing
- Sessions 6–9: Backend APIs – FastAPI, PostgreSQL, ORM, product management
- Sessions 10–11: Auth & roles – JWT login, admin vs customer
- Sessions 12–13: Cart & orders – checkout flow and order history
- Session 14: Admin dashboard & analytics
- Session 15: Production deployment
- Sessions 16–17: Shipper module (delivery workflow)



Thank You